

KARDZHALI, BulgariaWMO: **157300**Lat: **41.650N**Lon: **25.383E**Elev: **1086**StdP: **14.13**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	20.8	9.6	9.3	20.9	13.9	11.6	24.5	41.0	47.4	37.4	36.4	3.8	0	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.2	92.7	71.3	89.8	70.2	87.2	69.2	75.0	87.3	72.9	85.1	71.1	82.9	10.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	120.2	82.4	69.2	111.6	79.5	67.2	104.0	77.3	39.3	87.3	37.3	85.4	35.6	83.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.3	32.3	22.5	DB	9.8	98.2	4.5	4.3	6.6	101.4	4.0	103.9	1.4	106.3	-1.8	109.5	(4)
(5)			WB	9.2	76.5	4.2	3.2	6.2	78.8	3.7	80.7	1.4	82.5	-1.7	84.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	36.2	39.5	45.8	54.0	62.4	69.8	74.8	75.1	66.8	56.4	47.2	38.5	(6)	
	DBStd	15.20	8.32	8.62	7.45	6.47	5.47	5.27	4.57	4.52	5.93	7.02	7.66	8.56	(7)	
	HDD50	1470	430	302	170	34	1	0	0	0	0	24	141	368	(8)	
	HDD65	4343	892	713	594	331	113	16	1	50	275	536	821		(9)	
	CDD50	3530	3	9	41	155	386	594	768	778	503	224	56	13	(10)	
	CDD65	926	0	0	0	2	33	161	304	314	103	9	0	0	(11)	
	CDH74	8796	0	0	1	33	310	1401	2993	3108	869	80	1	0	(12)	
	CDH80	3352	0	0	0	3	44	444	1249	1340	260	12	0	0	(13)	
(14) Wind	WSAvg	6.6	5.4	7.5	8.6	7.7	6.6	5.7	7.2	6.8	5.9	5.8	5.9	6.0	(14)	
(15) Precipitation	PrecAvg	25.6	2.1	2.4	2.1	2.1	2.6	2.0	1.3	1.1	1.9	2.3	2.4	3.1	(15)	
	PrecMax	42.8	6.5	5.9	5.4	5.7	7.4	5.1	4.2	2.8	6.1	5.8	7.2	7.6	(16)	
	PrecMin	15.2	0.1	0.1	0.2	0.4	0.8	0.2	0.0	0.0	0.1	0.2	0.0	0.0	(17)	
	PrecStd	5.6	1.5	1.7	1.4	1.1	1.6	1.2	1.1	0.9	1.5	1.5	1.6	2.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	64.1	70.2	78.6	84.3	91.7	98.1	97.2	90.6	81.8	69.9	62.3	(19)	
		MCWB	51.5	52.2	56.4	64.9	66.2	72.3	72.4	70.9	68.8	64.2	58.5	56.0	(20)	
	2%	DB	55.3	59.6	65.4	72.8	80.5	87.9	92.9	93.1	86.5	75.5	65.5	58.6	(21)	
		MCWB	49.0	50.5	54.3	59.4	64.6	70.8	71.7	71.0	66.9	61.8	56.8	53.9	(22)	
	5%	DB	52.3	56.2	61.6	69.2	77.4	85.1	89.8	90.1	82.7	71.5	62.3	55.2	(23)	
		MCWB	47.1	49.3	52.2	57.2	63.7	69.5	70.3	70.2	65.4	59.9	56.0	50.6	(24)	
	10%	DB	48.9	53.0	58.1	65.7	74.4	82.1	86.8	87.4	79.1	67.8	59.1	51.2	(25)	
		MCWB	44.4	47.4	50.3	55.3	62.3	68.0	69.2	69.1	63.9	58.6	54.1	47.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	54.7	58.3	66.1	69.8	75.8	77.8	77.5	73.2	65.9	60.8	57.5	(27)	
		MCDB	56.8	60.0	66.2	76.8	80.0	87.4	90.7	90.0	87.8	76.1	66.0	60.6	(28)	
	2%	WB	50.2	52.0	55.6	61.5	66.8	73.4	75.0	74.9	68.8	63.7	58.8	54.4	(29)	
		MCDB	54.6	57.6	63.0	70.3	77.3	84.8	87.5	87.3	81.1	72.5	63.5	58.0	(30)	
	5%	WB	47.5	49.9	53.3	58.5	64.9	71.3	72.7	72.7	66.7	61.8	56.7	51.1	(31)	
		MCDB	51.5	55.7	59.9	67.0	74.8	81.5	84.9	85.3	78.7	68.9	61.4	54.7	(32)	
	10%	WB	44.8	47.5	51.2	56.3	63.1	69.3	70.7	70.6	65.1	59.8	54.4	48.0	(33)	
		MCDB	48.3	52.6	57.1	63.8	72.4	79.2	82.6	83.6	76.3	66.2	58.7	50.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	15.0	16.4	17.9	19.0	20.3	21.8	22.2	20.2	17.1	14.9	12.7	(35)	
		MCDBR	19.1	21.4	23.1	23.7	24.2	24.7	26.4	26.2	26.3	23.8	20.0	16.9	(36)	
	5% WB	MCWBR	12.8	13.4	13.4	12.5	11.4	10.4	10.2	10.2	11.2	11.8	12.1	11.6	(37)	
		MCWBR	16.4	19.7	20.6	21.7	21.9	22.4	23.7	23.4	23.4	21.0	17.0	14.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.340	0.373	0.422	0.478	0.459	0.461	0.459	0.466	0.443	0.402	0.368	0.346	(40)	
		taud	2.343	2.240	2.125	1.981	2.108	2.147	2.164	2.136	2.166	2.262	2.311	2.317	(41)	
	Ebn at Noon		252	259	259	252	261	260	259	253	249	246	238	237	(42)	
		Edh at Noon	27	35	44	55	49	48	47	46	42	34	28	26	(43)	
(44) All-Sky Solar Radiation	RadAvg		533	783	1122	1488	1848	2017	2124	1938	1428	945	610	468	(44)	
	RadStd		71	89	101	165	128	170	149	104	102	97	64	69	(45)	

Historical Trends

	DBAvg
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